

COLUMBIA OWENS, SC, USA

WMO: 723104

Lat: 33.971N

Lon: 80.996W

Elev: 212

StdP: 14.58

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 53867

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.3	28.3	5.1	7.2	34.8	9.8	9.1	37.6	19.8	56.9	17.9	55.3	1.4	320	0.356

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.7	97.3	75.2	94.9	74.9	92.8	74.5	78.5	89.8	77.8	88.7	77.0	87.4	7.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.6	134.9	81.7	75.0	132.5	81.3	74.2	128.9	80.8	42.1	90.3	41.2	88.8	40.4	87.5	83.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.4	13.0	11.2	20.0	100.7	3.3	3.3	17.6	103.1	15.7	105.0	13.8	106.9	11.4	109.3		
			17.5	80.5	3.1	1.3	15.3	81.5	13.5	82.2	11.7	83.0	9.5	83.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	65.4	46.5	50.1	57.1	65.3	73.1	80.1	82.5	81.8	76.6	66.1	55.5	49.5
	DBStd	14.67	9.83	8.69	8.65	7.33	5.89	4.14	3.54	3.73	5.24	7.45	8.05	8.98
	HDD50	482	190	99	34	2	0	0	0	0	0	1	34	122
	HDD65	2229	576	423	270	86	10	0	0	0	2	78	300	484
	CDD50	6115	81	101	254	461	716	902	1008	986	799	501	198	108
	CDD65	2389	3	5	25	95	261	452	543	521	351	113	15	5
	CDH74	23490	18	37	260	972	2440	4549	5865	5378	3009	842	99	21
	CDH80	10281	1	2	43	283	959	2156	2872	2514	1230	214	7	0
Wind	WSAvg	3.7	4.2	4.4	4.7	4.7	4.1	3.7	3.5	2.9	3.1	2.9	2.9	3.5
Precipitation	PrecAvg	46.1	4.2	3.8	4.6	3.0	3.2	5.0	5.4	4.5	3.8	2.8	2.8	3.1
	PrecMax	66.0	9.4	8.6	10.0	7.3	7.4	12.6	14.9	10.4	10.2	11.9	7.4	9.8
	PrecMin	30.6	0.7	0.6	0.5	0.2	0.7	0.9	1.0	0.8	0.0	0.0	0.4	0.4
	PrecStd	7.6	2.1	1.9	2.3	1.9	1.6	2.7	3.2	2.7	2.4	2.5	1.7	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.4	79.0	84.3	90.0	95.5	99.4	100.2	99.9	95.1	89.3	81.2	76.8
		MCWB	64.5	64.5	64.4	68.2	71.0	74.6	76.6	75.8	72.3	73.9	67.0	65.5
	2%	DB	71.1	73.6	80.7	86.3	91.3	95.8	97.4	96.6	92.6	85.3	77.0	72.5
		MCWB	61.2	60.6	63.2	66.7	70.4	74.2	76.1	75.8	72.6	69.7	64.0	64.0
	5%	DB	66.8	69.9	76.6	83.5	88.8	93.0	94.8	93.5	90.2	82.0	73.2	68.5
		MCWB	58.7	58.3	60.9	65.5	70.0	74.1	75.7	75.6	72.3	68.2	62.3	62.2
	10%	DB	62.9	65.7	72.5	80.4	85.6	90.4	92.2	91.1	87.5	78.9	70.0	64.8
		MCWB	56.0	56.6	58.9	64.0	69.0	73.4	75.5	75.0	72.0	67.2	61.6	59.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.8	67.1	68.2	73.0	75.7	78.4	80.3	79.9	77.6	76.3	71.0	69.7
		MCDB	73.1	73.0	76.7	84.9	86.1	90.3	92.6	91.1	86.7	83.1	75.1	73.5
	2%	WB	63.8	64.3	65.9	70.7	73.9	77.2	78.8	78.6	76.4	74.4	68.1	66.3
		MCDB	68.7	71.0	75.1	80.5	83.5	88.9	90.8	89.3	84.9	80.6	72.9	69.9
	5%	WB	60.8	61.4	64.1	68.8	72.6	76.3	77.9	77.8	75.4	72.5	65.9	63.6
		MCDB	65.4	67.9	72.6	77.7	82.1	87.2	89.4	88.1	83.4	78.4	70.5	67.8
	10%	WB	57.2	58.2	62.0	66.7	71.3	75.4	77.1	77.0	74.3	70.0	63.3	59.9
		MCDB	61.4	64.0	69.5	75.4	81.0	85.5	88.0	86.6	81.8	76.0	68.7	64.5
Mean Daily Temperature Range	5% DB	MDBR	19.9	20.8	22.3	22.7	20.9	19.1	18.7	17.8	18.2	20.4	21.9	19.6
		MCDBR	24.1	25.4	26.7	26.3	24.7	22.3	21.7	21.2	21.7	22.5	24.8	21.9
		MCWBR	15.5	14.9	12.1	10.1	8.2	6.3	5.9	5.5	6.2	9.6	13.8	14.5
	5% WB	MCDBR	20.2	21.4	21.8	20.6	19.3	19.0	18.8	18.1	16.8	15.8	18.0	19.2
		MCWBR	15.5	15.0	12.3	10.2	7.4	6.3	6.3	5.6	6.0	8.3	12.5	15.0
Clear-Sky Solar Irradiance	taub	0.324	0.333	0.371	0.412	0.477	0.510	0.556	0.528	0.444	0.402	0.351	0.321	
	taud	2.489	2.461	2.382	2.259	2.125	2.075	1.969	2.038	2.257	2.342	2.444	2.533	
	Ebn at Noon	281	291	286	278	260	251	238	243	259	262	268	276	
	Edh at Noon	27	31	36	43	50	52	58	53	41	35	28	25	
All-Sky Solar Radiation	RadAvg	880	1080	1422	1789	1949	2029	1928	1755	1511	1263	967	767	
	RadStd	46	114	106	130	152	142	126	87	130	151	76	71	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.42	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+587	+340
(47)	Regional (62 neighbors)	+1.06	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+340	+239

Nomenclature: See separate page